

Volatility Models

(2. The Dupire Local Volatility Model)

Local Volatility

$$ds = rSdt + \sigma S^\beta dW$$

$$ds/s = rdt + \sigma S^{\beta-1} dW$$

$$\frac{dS(t)}{S(t)} = rdt + \underbrace{\sigma_L(t, S)}_{\text{(local volatility)}} dW(t)$$

✓
Parametric method

$$\sigma_L(t, S) = \sigma S^{\beta-1} \rightarrow (\text{CEV})$$

$$\beta = 1 \quad (\text{Black-Scholes})$$

$$\beta = 0 \quad (\text{Bachelier Model})$$

$$\beta = 1/2 \quad (\text{square-root process})$$

× Cannot match the surface perfectly

Non-parametric method

$$\left[\sigma(t, S) \right]$$

(Dupire Local volatility model)

✓ Can match the surface perfectly ✓
× Does not capture any volatility dynamics



$$dr = \theta(t) dt + \sigma dW$$

Strike Delta and Gamma (Dual)

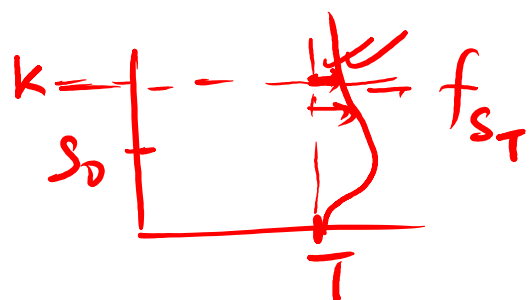
Dual deltas are defined as first order sensitivity of option to strike price. It may sound illogical to talk about changes in strike price as strike price remains constant for the contract. However, this can be interpreted as the difference in price of another option whose strike is slightly different than the first.

The Strike deltas are given as

$$\frac{\partial c}{\partial K} = -e^{-rT} N(d_2) \quad \frac{\partial p}{\partial K} = e^{-rT} N(-d_2)$$

Since $N(d_2)$ is the risk-neutral probability of exercising the call option, the strike deltas or dual deltas simply give the discounted risk-neutral probability.

The risk-neutral density of the stock can be retrieved by differentiating the option price twice. In other words, the strike gamma gives the risk-neutral density of the underlying price. (Breeden-Litzenberger)



$$\frac{\partial^2 c}{\partial K^2} = e^{-rT} f_{S_T}(K)$$

Strike Delta and Gamma (Dual)

Derive the two formulas

$$✓ \quad \frac{\partial c}{\partial K} = -e^{-rT} N(d_2)$$

$$✓ \quad \frac{\partial^2 c}{\partial K^2} = e^{-rT} f_{ST}(K)$$

Hint : Use Leibniz Rule on call price as expectation

Leibniz Rule

$$\frac{d}{dx} \left(\int_{a(x)}^{b(x)} f(x, t) dt \right) = \underline{f(x, b(x))} \cdot \underline{\frac{db(x)}{dx}} - \underline{f(x, a(x))} \cdot \underline{\frac{da(x)}{dx}} + \int_{a(x)}^{b(x)} \frac{\partial}{\partial x} f(x, t) dt$$

The Dupire formula

Show the following identities

$$\checkmark \quad \underline{\underline{C(S_t, t; K, T)}} = \underline{\underline{S_t}} \underline{\underline{\frac{\partial C}{\partial S_t}}} + K \underline{\underline{\frac{\partial C}{\partial K}}}$$

$$\checkmark \quad \underline{\underline{S_t^2 \frac{\partial^2 C}{\partial S_t^2}}} = K^2 \underline{\underline{\frac{\partial^2 C}{\partial K^2}}}$$

Substituting these identities in Black Scholes PDE,

$$\begin{aligned} & \frac{\partial C}{\partial t} + \frac{1}{2} \sigma^2 S_t^2 \frac{\partial^2 C}{\partial S_t^2} + \underbrace{rS_t}_{\checkmark \underline{\underline{V(S,t)}}} \frac{\partial C}{\partial S_t} - \underbrace{rC}_{\textcircled{+}} = 0 \\ \Rightarrow & \left(\frac{\partial C}{\partial t} \right) + \frac{1}{2} \sigma^2 K^2 \frac{\partial^2 C}{\partial K^2} - rK \frac{\partial C}{\partial K} = 0 \\ \Rightarrow & -\frac{\partial C}{\partial T} + \frac{1}{2} \sigma^2 K^2 \frac{\partial^2 C}{\partial K^2} - rK \frac{\partial C}{\partial K} = 0 \end{aligned}$$

$$\underline{\underline{\sigma(T, K)}} = \sqrt{\frac{\frac{\partial C}{\partial T} + rK \frac{\partial C}{\partial K}}{\frac{1}{2} K^2 \frac{\partial^2 C}{\partial K^2}}}$$

Dupire Formula

Duality



dr

Black-Scholes PDE

$$\frac{\partial C}{\partial t} + \frac{1}{2} \sigma(S, t)^2 S^2 \frac{\partial^2 C}{\partial S_t^2} + rS \frac{\partial C}{\partial S_t} - rC = 0$$

K is fixed, S diffuses

t, S

Dupire PDE

$$-\frac{\partial C}{\partial T} + \frac{1}{2} \sigma(K, T)^2 K^2 \frac{\partial^2 C}{\partial K^2} - rK \frac{\partial C}{\partial K} = 0$$

S is fixed (current value), K diffuses

t, S

Calibrating the Dupire vol

SVI, CSVI, eSVI

The local volatility $\sigma(K, T)|_{T=t, K=S}$ can be derived from Dupire's formula by taking finite differences of option prices to approximate the derivatives.

$C = a + bK + dK^2$
 $\frac{\partial C}{\partial K} = b + 2dK$

$\Delta T_j \rightarrow$ K, T

	T_1	T_2	T_3	T_4
$\Delta K_i \downarrow$				
K_1				
K_2				
K_3		$C(K_i, T_j)$		
K_4				
K_5				

$$\frac{\partial C}{\partial T} = \frac{C(K_3, T_3) - C(K_3, T_2)}{T_3 - T_2}$$

	T_1	T_2	T_3	T_4
K_1				
K_2				
K_3		$\sigma(K_i, T_j)$		
K_4				
K_5				

$$\sigma(T, K) = \sqrt{\frac{\frac{\partial C}{\partial T} + rK \frac{\partial C}{\partial K}}{\frac{1}{2} K^2 \frac{\partial^2 C}{\partial K^2}}}$$

wings

Finite difference approximations

$$\frac{\partial C}{\partial T} = \frac{C(T + \Delta T, K) - C(T, K)}{\Delta T}$$

$$\frac{\partial C}{\partial K} = \frac{C(T, K + \Delta K) - C(T, K)}{\Delta K}$$

$$\frac{\partial^2 C}{\partial K^2} = \frac{C(T, K + \Delta K) + C(T, K - \Delta K) - 2C(T, K)}{\Delta K^2}$$

Once local volatility is calibrated, we could price other options using Monte Carlo or Finite Difference Methods

Implied volatility to Local volatility

For each volatility term, we need four option values to evaluate the expression. The denominator is the density of stock price for which finite difference approximations can lead to significant errors for high strike prices. (density is close to 0). Local volatility will come out unrealistically large. The improvement suggested in literature is to use smooth arbitrage free surface first and then differentiate analytically.

It's useful to work from implied volatility than prices which gives more stable estimates. Here is the modified formula for local vol from IV surface

$P \rightarrow IV$
 $C(K, T) \rightarrow \sigma_{imp}(K, T)$

$$y = \log\left(\frac{K}{S_F}\right) = \log\left(\frac{K}{S_0}\right) - rT$$

$$w = \sigma_{imp}^2(T, K)T$$

$f = S e^{rT}$

$$\sigma_{loc}^2(T, K) = \frac{\frac{\partial w}{\partial T} + rK \frac{\partial w}{\partial K}}{1 + K \frac{\partial w}{\partial K} \left(\frac{1}{2} - \frac{y}{w}\right) + \frac{1}{2} K^2 \frac{\partial^2 w}{\partial K^2} + \frac{1}{2} K^2 \left(\frac{\partial w}{\partial K}\right)^2 \left(-\frac{1}{8} - \frac{1}{2w} + \frac{y^2}{2w^2}\right)}$$

$$\frac{\partial w}{\partial T} = \sigma_{imp}^2 + 2T\sigma_{imp} \frac{\partial \sigma_{imp}}{\partial T}$$

$$\frac{\partial w}{\partial K} = 2T\sigma_{imp} \frac{\partial \sigma_{imp}}{\partial K}$$

$$\frac{\partial^2 w}{\partial K^2} = 2T \left(\frac{\partial \sigma_{imp}}{\partial K}\right)^2 + 2T\sigma_{imp} \frac{\partial^2 \sigma_{imp}}{\partial K^2}$$